

$\therefore$ By Defn. 3, there exists an element $\{x\}$ of $\mathcal{P}(A)$. But $\because x \notin B$.

$\therefore$ There exists no set containing x which is a member of $\mathcal{P}(B)$.

So, $\mathcal{P}(A) \neq \mathcal{P}(B)$. This is a contradiction.

$$\boxed{\therefore A=B} \quad \square$$

25) How many elements does each of these sets have where a and b are distinct elements?

a) $\mathcal{P}(\{a, b, \{a, b\}\})$
 $= \{\emptyset, \{a\}, \{b\}, \{a, b\}, \{\{a, b\}\}, \{a, \{a, b\}\}, \{b, \{a, b\}\}, \{a, b, \{a, b\}\}\}$

Cardinality: 8

b) $\mathcal{P}(\{\emptyset, a, \{a\}, \{\{a\}\}\})$
 $= \{\emptyset, \{\emptyset\}, \{a\}, \{\{a\}\}, \{\emptyset, a\}, \{\emptyset, \{a\}\}, \{\emptyset, \{\{a\}\}\}, \{a, \{a\}\}, \{a, \{\{a\}\}\}, \{\{a\}, \{\{a\}\}\}, \{\emptyset, a, \{a\}\}, \{\emptyset, a, \{\{a\}\}\}, \{a, \{a\}, \{\{a\}\}\}, \{\emptyset, \{a\}, \{\{a\}\}\}, \{\emptyset, a, \{a\}, \{\{a\}\}\}\}$

Cardinality: 16

c) $\mathcal{P}(\mathcal{P}(\emptyset))$ $\mathcal{P}(\emptyset) = \{\emptyset\}$ $\mathcal{P}(\mathcal{P}(\emptyset)) = \{\emptyset, \{\emptyset\}\}$

Cardinality: 2

26) Determine whether each of these sets is the power set of a set, where a and b are distinct elements.

a) $\emptyset$. We know that a set of n elements has 2^n elements in its power set. $2^n = 0$. There exist no such n .

$\therefore \emptyset$ is not a power set.

b) $\{\emptyset, \{a\}\}$. The power set of $\{a\}$ is: $\{\emptyset, \{a\}\}$

c) $\{\emptyset, \{a\}, \{\emptyset, a\}\}$. The cardinality is 3. $2^n = 3$. There exists no such integer n . $\therefore$ This is not a power set.

d) $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. The power set of $\{a, b\}$ is: $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$. $\square$